



OPTIMIZING GATE ASSIGNMENTS AT AIRPORT TERMINALS

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Abstract—The assignment of aircraft arriving on schedule to available gates is a major issue during the daily airline operations. The Gate Assignment Problem (GAP) is an easily-understood but difficult to solve problem. An important issue in GAP is to minimize the distance a passenger is required to walk inside the terminal to reach his or her departure gate. The total passenger walking distance is based on the passenger transfer volume between every pair of aircrafts and the distance between every pair of gates. Therefore, the problem of assigning gates to arriving and departing flights at an airport is a Quadratic Assignment Problem (QAP), which is a difficult problem to solve. Recently, integer programming (IP), simulation, and expert systems have been applied to the GAP. A new integer programming formulation of the gate assignment problem is presented. An efficient heuristic solution procedure for solving the problem is also proposed. The performance of this solution approach is evaluated in a variety of test problems, and is compared to the optimal solutions obtained using CPLEX in certain test problems. The proposed heuristic solution approach proves to be an efficient method for obtaining good solutions for large scale gate assignment problems in a very reasonable computation time. © 1998 Elsevier Science Ltd. All rights reserved

Keywords: gate assignment, quadratic assignment, walking distance, baggage transfer, apron time

1. INTRODUCTION

1.1. Background

The distance a passenger is required to walk in an airport to reach his departure gate, the baggage claim area, or his connecting flight is an important consideration in effective utilization of aircraft gates at an airport terminal (Mangoubi and Mathaisel, 1985). These distances can be reduced by improving the method by which scheduled flights are assigned to the airport terminal gates. In 1977 Braaksma demonstrated that the procedure can have a profound impact on passenger walking distances without changing the layout of the terminal area. As an example, the mean walking distance per passenger at Terminal No. 2 of Toronto International Airport in Canada was reduced from 923 ft in 1973 to 800 ft in 1975. This improvement was a direct result of a change in gate assignment policy by Air Canada, the terminal's sole user. The decision represented a savings of over 100 ft per passenger.

Therefore, an important question is which flight-to-gate assignment policy results in the minimum walking distances. The challenge is to develop an approach to the problem which yields the minimum total, or average walking distance per passenger based on actual or pre-planned time tables of flight schedules. For the purpose of planning, the flight time tables should be assumed fixed and given, as should be the airport terminal layout. No changes should be required in the flight schedule or the airport terminal.

1.2. Problem statement

Gate assignment is a complex process. Basically, it may take into account the following factors (Chang, 1994):

1. Walking distances for transfer, terminating, and originating passengers;
2. Baggage handling distance for transfer, terminating, and originating passengers;
3. Time tables of flight schedules; and,
4. Aircraft-gate size compatibility.

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To minimize the time that transfer passengers need to walk from one gate to another, it would be the best to assign connecting flights at nearby or the same gates if the flight schedules are not conflicting. However, one cannot overlook the importance of the originating and terminating passengers who play an important role in an airport. In the past, several researchers have worked on this problem, but most have focused on solving the minimum total passenger walking distance during one time period or for a batch of aircraft, which means that the gate assignment has to be accomplished in each time period. In this way, no new flight can be assigned to a gate before the end of the previous time period as shown on Fig. 1(a). In this case, a batch of aircrafts are assumed to arrive and depart within a short time 'slot', from time T_a to Time T_d in Fig. 1(a), and all aircrafts are on the ground simultaneously for at least a short period so that transfers can be made. In this situation, it is clear that the average gate utilization will be lower for a long time period.

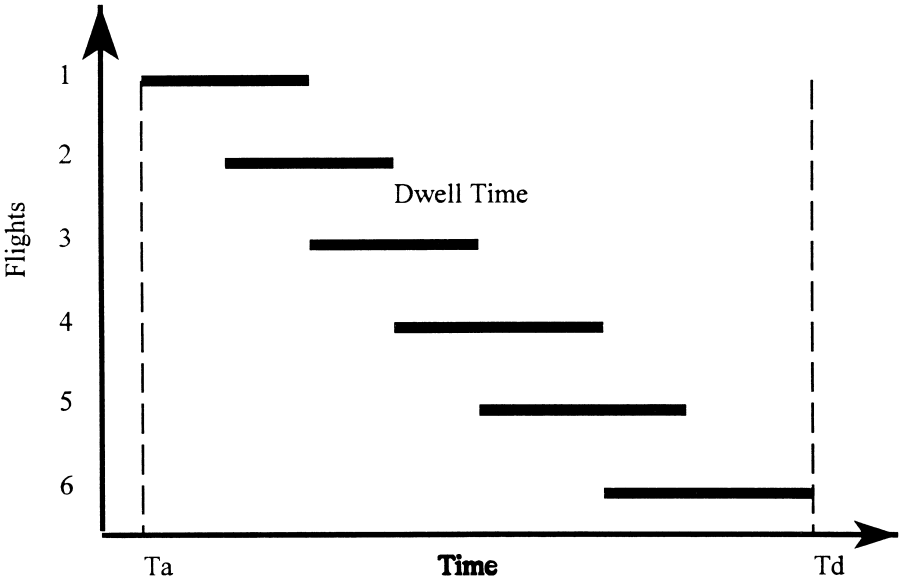
There are several reasons for dealing with the gate assignment problem in one time period. One reason is that the gate shortages are only of concern during the peak hours and we try to assign available gates to the flights during the peak hours. When proper gate assignment is done for the peak hours, there won't be a gates shortage problem outside the peak hours. The other reason is that it is difficult to deal with the flight schedules which are overlapping on two continuous time 'slots', which involves dealing with a multi-slot gate assignment problem using conventional gate assignment approaches. Size compatibility of the gate and aircraft as well as gate's access to government inspection service are also important. A large gate has the flexibility to accommodate various sizes of aircraft whereas a small gate is more limited. International aircraft should park at gates with easy access to governmental inspection services. Passenger transfer flows are highly related to the demand generated by each origin and destination. In an attempt to reduce cost, airlines positioning of large aircraft can have major effects on the overall gate assignment efficiency. Gate assignment should consider the overall passenger walking distance and baggage handling distance as well as ground time periods and dwell durations.

In addition, modern airport terminal building designs with finger piers and air bridges sometimes result in significant walking distances for passengers. This comprises the distance between check-in desks and gates with air bridges for departing passengers, the distance between gates and the baggage claim area for arriving passengers, and the distance between gates for transferring passengers.

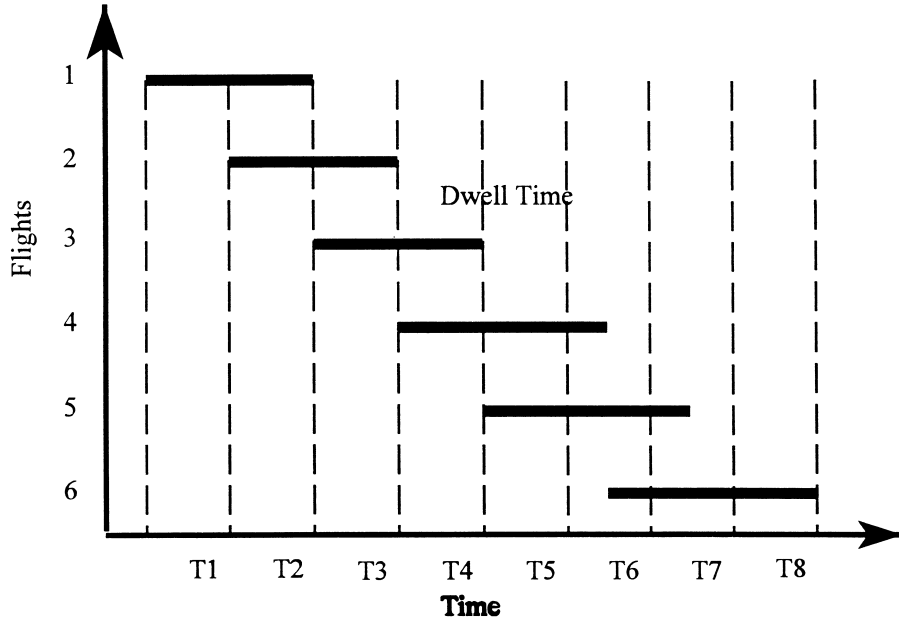
Therefore, for a given terminal building configuration, when the distances between all gates and gates to check-out or check-in point are known, the average passenger walking distance may vary depending on the particular assignment of aircraft to aircraft stands. This happens when the number of passengers vary between aircrafts, which is a common situation. The optimum strategy for one-time cross section is obvious: aircraft carrying more passengers should be parked closer to the other aircraft which also carry more passengers. However, finding the optimum strategy for a given period of time, which also takes into account aircraft stand occupancy time, is a more complex (dynamic) problem that is combinatorial by nature.

In this paper, we formulate a multi-slot [or several time periods as shown on Fig. 1(b)] gate assignment problem as an integer program. Then, we employ branch and bound technique to solve the problem and obtain the optimal solutions. Also, a new heuristic approach is proposed in this study and the results of its implementation are compared with the optimal solutions for the gate assignment problem. The new heuristic algorithm can effectively assign aircraft to gates to minimize total passenger walking distances based on the time table of flight schedules.

There are six sections in this paper. The existing literature are briefly reviewed in Section 2. The integer programming formulation and the heuristic solution algorithm for the gate assignment problem are discussed in Sections 3 and 4. Several applications of the proposed heuristic algorithm to solve the gate assignment problem are presented in Section 5. Nine test problems from the literature and 33 test problems generated in this study are used to verify the proposed algorithm. Also, we solve the gate assignment problem using a commercially available linear and integer programming software, CPLEX, and compare the quality of the solutions from the heuristic algorithm with the optimal solutions. The gate assignment problem is an NP-hard problem (Obata, 1979). Even for a moderate size problem the optimal solution cannot be found in a reasonable execution time. Finally, Section 6 provides the summary and conclusions, and discusses the future research directions.



(a) One Ground Time Period



(b) Several Ground Time Periods

Fig. 1. Aircraft activities at an airport.

2. LITERATURE REVIEW

Passenger walking distance is of major importance when design alternatives for a terminal are evaluated. One of the first efforts to try to incorporate a quantitative method to minimize intra terminal travel into the design process was presented by Braaksma and Shortreed (1975). Wirasinghe *et al.* (1987) present a method to calculate walking distance for finger-pier terminal

geometry as well as a method to optimize that geometry due to walking distance. Robuste (1988) proposed a similar type of model that addressed primarily hub transfer of both passengers and baggage. Bandara (1989) extended the work presented by Wirasinghe *et al.* (1987), introducing more terminal types, hub transfers, unequal piers, etc. and offering some general guidelines in the area of preliminary terminal design. The suitability of particular types of terminals for certain traffic volume and passenger portioning is recommended. These results are reported in Bandara and Wirasinghe (1992).

The Gate Assignment Problem (GAP) is an easily-understood but difficult to solve problem. Integer programming (IP), simulation, and expert systems have been applied to the GAP. As IP stresses problem formulation and solution algorithm development, most of literature handled the GAP which minimizes passenger walking distance for a batch of arriving aircrafts at the same time. Simulation models can consider the stochastic situation at the airport and handle uncertain factors. The expert systems emphasize the integration of the rules and models. However, those rules and models depend on the IP and simulation studies. The basic constraints of the IP are that one aircraft is to be assigned to only one gate and two aircraft cannot be assigned to the same gate when their apron times overlap. These are well defined and contribute to the realistic nature of the model. All gates are assumed to be used by all the aircraft types considered, and both the number of arriving and departing passengers and flight arrival and departure time are known in advance.

Babic *et al.* (1984) formulated the GAP as a linear 0–1 IP. A branch-and-bound algorithm is used to find the optimal solution. Transfer passengers are not considered in this paper. The optimal strategy is that aircraft carrying more passengers should be assigned to the gate closer to the central part of the building. Similarly, Mangoubi and Mathaisel (1985) used linear programming (LP) relaxation of an IP formulation and greedy heuristic approaches to solve the previous GAP (Babic *et al.*, 1984). The distances for arriving and departing passengers are defined similar to the previous case, but the difference is that this model includes transfer passengers as well. However, the transfer passengers and their walking distances are not computed as precisely as for local passengers.

Airport Forum (1986) reports on the SAS application of ARGUS, a computerized ramp and gate utilization system at Copenhagen Kastrup Airport. The goal of this application was to minimize transfer passenger walking distances along finger piers.

Bihl (1990) used 0–1 LP to solve the minimum walking distance GAP for fixed arrivals in a hub operation. This GAP is formulated as an ‘assignment problem’ which is more easily solved than the above two GAPs. The results of these three papers indicate that a significant improvement in walking distance is obtained.

Simulation analysis is used by Baron (1969) to analyze the effects of passenger walking distanced resulting from different gate-use strategies. Both local and transfer passengers are considered.

Since traditional operations research approaches have difficulty with uncertain information and multiple criteria, expert systems offer some advantages over conventional approaches. Hamzawi (1986) developed a microcomputer-based gate assignment model as a planning and operational tool for simulating the assignment of gates to aircraft and evaluating the effectiveness of operational options to improve the gate utilization. Gosling (1990) used an expert system to solve a GAP at a major airline hub (Denver Stapleton Airport). Srihari and Muthukrishnan (1991) applied similar expert systems to solve the aircraft-gate assignment problem. The significant difference between these two expert systems is that Srihari and Muthukrishnan’s model has a module to do sensitivity analysis.

Although baggage transport distance has not been a criterion for the GAP before Chang’s study (1994), baggage transport distance is an important criterion in the hub operations and should be considered in the GAP. Robuste and Daganzo (1992) studied the containerized baggage handling process in hub operations.

The GAP at an airport is a Quadratic Assignment Problem (QAP). The QAP is a location problem where the cost associated with placing a facility (i.e. aircraft in this study) at a certain site (i.e. gate in this study) depends not only on the distance from other facilities and the demands, but also on the interaction with other facilities. Similarly, the GAP at a hub is a problem of minimizing the total local passenger walking distance (and their baggage distance) as well as the total

transfer passenger walking distance (and their baggage transport distance). The total passenger walking distance is associated with the inter-flow between every pair of aircraft and the distance between every pair of gates. Consequently, the gate assignment problem is a QAP.

Since QAPs are NP-hard (there are no polynomial time bound algorithms for their solution), only implicit enumeration methods are known for solving them optimally. Essentially, there are three different types of algorithms that have been used for implicit enumeration, such as branch-and-bound type algorithms, cutting plan methods, and dynamic programming.

Koopmans and Beckman (1957) were the first to formulate a mathematical model for locating facilities for this situation. The model, now known as quadratic assignment problem is: given a set $N = \{1, 2, \dots, n\}$ and real numbers c_{ik} , a_{ik} , b_{ik} for $i, k = 1, 2, \dots, n$, find a permutation of the set N which minimizes:

$$Z = \sum_{i=1}^n c_{i\varphi(i)} + \sum_{i=1}^n \sum_{k=1}^n a_{ik} b_{\varphi(i), \varphi(k)} \quad (1)$$

Later this problem was extended to more general QAP as introduced by Lawler (1963) given real numbers $d_{ijkl}(i, j, k, l \in N)$, find a permutation $\varphi: N \Rightarrow N$ which minimizes:

$$Z = \sum_{i=1}^n \sum_{k=1}^n d_{i, \varphi(i), k, \varphi(k)} \quad (2)$$

There have been many other proposed applications of QAPs, such as in building layout, machine scheduling and keyboard arrangement. Obata (1979) systematically reviewed the formulations of QAPs and solution methods. In addition, he proposed some new algorithms for the case where the set of feasible sites is given by a rectangular array.

Burkard and Derigs (1980) presented two heuristic methods: the method of increasing degrees of freedom, and the cutting plan and exchange method. Suryanarayanan (1989) developed an algorithm called Linear Placement with Multidimensional Scaling (LPMDs) to solve the Linear Placement Problem which is a one dimensional QAP. Her algorithm makes use of the ordering nature of the Multidimensional Scaling technique to construct a starting solution. This, when combined with the Los (1978) pair-wise exchange algorithm, performs remarkably well on the problems even up to size 40.

The QAP is discussed in detail by Mirchandani and Francis (1990). They presented various integer programming formulations of QAPs and studied their computational complexity. In addition, the current powerful exact and heuristic QAP solution algorithms are introduced clearly.

Suryanarayanan *et al.* (1991) proposed a hybrid algorithm which combines the Los pair-wise exchange algorithm and Multidimensional Scaling to solve the Linear Placement Problem remarkably well.

Chang (1994) formulated the GAP as a QAP to minimize the walking distance and baggage transferring distances in airport hub operations. He proposed an accessibility value sorting algorithm to solve the GAP. Then, this algorithm is improved by either Los pair-wise or Obata mixed exchange algorithm into the Sorting-Los and Sorting-Mixed algorithms respectively. He claims that the Sorting-Mix algorithm is better than the Sorting-Los algorithm in terms of solution accuracy and execution time. In addition, Chang also developed a Simulated Annealing algorithm to solve the GAP. Finally, he concluded that Simulated Annealing is better than the Sorting-Los and Sorting-Mix algorithms in terms of solution accuracy.

Although the GAP has been formulated as QAP, and QAP has been solved by several good algorithms in the literature, the time constraints have not yet been considered in quadratic assignment problems. This study extends the quadratic assignment problem formulations by introducing time constraints in a gate assignment problem.

3. PROBLEM FORMULATION

The assignment of arriving aircrafts to available gates is one of major issues in airport operations. The use of hub and spoke networks (HSNs) in airline operations and the consequent increase in the volumes of passengers and baggage that must be redistributed among flights has

significantly increased the complexity of gate assignment. Inefficient gate assignment increases costs, worsens airline service quality and lengthens passenger transfer delay.

Passenger walking distance is a major criterion for evaluating gate assignment. While walking distance affects passengers, the baggage transfer distance affects the baggage handling costs of airlines. Since gates may have different characteristics in the same terminal, the flight schedules (gate occupancy time or flight dwell time) must also be taken into account. Due to the complexity of airport operations and the difference in walking distance between local passengers and transfer passengers, it is difficult to solve the minimum Passenger distance Gate Assignment Problem (PGAP) which involve the time tables of flight schedule.

The complexity of gate assignment at a hub is greater than that at a spoke city. According to the characteristics of the hub operations, passengers are divided into two major groups: local passengers and transfer passengers (Chang, 1994).

1. Local passengers are originating or terminating passengers:
 - (a) Originating (departing) passengers are those passengers arriving at the terminal by ground transportation. They walk from ground transportation facilities to the departure gate through security checks and, when necessary, check-in counters.
 - (b) Terminating (arriving) passengers are those arriving at the terminal by aircraft. They usually walk from the arrival gate to ground transportation facilities through, when necessary, the baggage claim area.
2. Transferring passengers transfer from one aircraft to another aircraft without leaving the terminal.

The following two costs are associated with the GAP:

1. Passenger walking: a passenger is required to walk inside the terminal to reach his departure gate, baggage claim area, or connecting flight.
2. Baggage transport: baggage must be transported from aircraft to terminal or to another aircraft. All bags must be sorted before they are transported to their departure aircraft.

The passenger walking distance and baggage transport distance may be affected by the configuration of the terminal, e.g. whether centralized or decentralized. Whatever the types of terminals, the GAP can be solved if the passenger walking and baggage transport distances are given.

Unlike baggage transport distance, walking distance directly impacts passengers. For customer satisfaction, it is important for airlines to minimize their passenger walking distances. Baggage processing, on the other hand, does not directly impact passengers as long as baggage is delivered to the passengers at the right airport. Besides, baggage is usually transported by vehicles outside the terminals. Passenger walking distance is usually a more important factor than baggage transport distance. This explains why the conventional GAP focuses on minimizing passenger walking distance rather than baggage transport distance. However, baggage transport distance affects the length of time which the flight has to remain at a gate, so it is important in terms of operating costs.

In order to measure the impact of passenger walking distances and flight schedules, this PGAP is formulated as a new nonlinear (quadratic) 0–1 IP problem which is a QAP. Rather than solving this problem exactly, a heuristic approach is proposed to solve the PGAP at airports.

Using a similar formulation the minimum baggage transport distance GAP (BGAP) is also formulated as a QAP (Chang, 1994). The combined minimum passenger distance and baggage distance GAP (PBGAP) is a QAP, too. When the algorithms developed to solve the minimum PGAP, they can also be used to solve the minimum PBGAP. Therefore, we focus on solving PGAP instead of PBGAP in this study. In our approach, the transfer of the passengers between terminals can easily be handled as long as the walking distance for such transfers are known.

Two approaches to the GAP are presented to solve the PGAP. One makes use of a branch-and-bound technique to solve an IP formulation, and the other employs a heuristic approach. The heuristic is interesting because it provides a good solution and computation time to walking distance problems, and its results might be used as an initial solution to the IP formulation if an optimal solution is desired.

In our modeling approach, we divide the study period into several time intervals. The length of the time intervals can vary, however, it should be selected very carefully because it has significant impact on the size of the problem to be solved. The length of the time interval should not be significantly shorter or longer than the minimum time that an aircraft can occupy a gate. Based on our experience, time interval lengths of less than 15 min are not very useful because the aircrafts usually occupy the gates more than this time. On the other hand, time intervals more than one hour seem to be impractical because such a long time interval does not allow effective gate utilization over time.

The aircraft arrival or departure times in this study are rounded to the arrival or departure time period, so each aircraft may dwell in a gate for several time intervals between its arrival and departure. The aircraft is assumed to occupy the gate from the beginning of the time period in which it arrives to the end of the time period in which it departs.

The aircraft-gate compatibility issue is accounted for by the in which the decision variables in this model are defined. Basically, when a gate cannot accommodate a particular type of aircraft at certain times, the corresponding decision variables are set equal to zero. This can be accomplished by pre-processing the decision variables and has no effect in the solution approach or the required computational time.

A binary variable X_{ijt} is defined for each possible flight-to-gate assignment in each time period:

$$X_{ijt} = \begin{cases} 1 & \text{if flight } i \text{ is assigned to gate } j \text{ in time period } t \\ 0 & \text{otherwise} \end{cases}$$

In addition, the terms P_{ii} , P_{io} , and P_{oi} will denote the estimated number of transferring passengers from flight I to flight I' , the number of arriving passengers from flight I to entrance, and the number of departing passengers from entrance to flight I respectively. The walking distance is denoted by $D_{jj'}$, D_{jo} , and D_{oj} , respectively, for a transfer passenger from gate j to gate j' , for an arriving passenger from gate j to entrance o , and for a departing passenger from entrance o to gate j . Figure 2 shows a simple terminal configuration.

Therefore, the assignment problem is expressed as:

$$\text{MIN} \sum_i \sum_{i'} \sum_j \sum_{j'} P_{ii'} D_{jj'} X_{ijt} X_{i'jt'} + \sum_i \sum_j (P_{io} D_{jo} + P_{oi} D_{oj}) X_{ijt} \quad (3)$$

The objective function consists of two parts: one is the total passenger walking distances of transfer passengers over all time periods. The other is the total passenger walking distances of local passengers over all time periods. The first part is a QAP. The second part is a LAP. The LAP is a location problem where the cost associated with placing a facility at a certain site depends only on the distance between facility and location. However, we can redefine

$$P_{ii'} D_{jj'} \leftarrow P_{io} D_{jo} + P_{oi} D_{oj} \quad \forall i, j \quad (4)$$

where, the entrance o is assumed as an available gate to be used for the arrival passengers and departure passengers. Then, we can deal with this gate assignment problem as a purely transfer passenger problem and obtain a purely quadratic objective function with cost coefficients $P_{ii'}, D_{jj'}$ as

$$\text{MIN} \sum_i \sum_{i'} \sum_j \sum_{j'} P_{ii'} D_{jj'} X_{ijt} X_{i'jt'} \quad (5)$$

This is an IP formulation of a QAP with a quadratic objective function; hence the term ‘quadratic assignment problem.’ Since linear problems are easier to solve than those with a quadratic objective function, we try to get rid of the quadratic terms in the objective function. We propose to replace the products $X_{ijt} X_{i'jt'}$ by n^4 new binary variables $Y_{ij'jt'}$:

$$Y_{ij'jt'} \equiv X_{ijt} X_{i'jt'} \quad (6)$$

where we define that a binary decision variable $Y_{ij'jt'}$ is assigned for each possible pair flight-to-gate assignment:

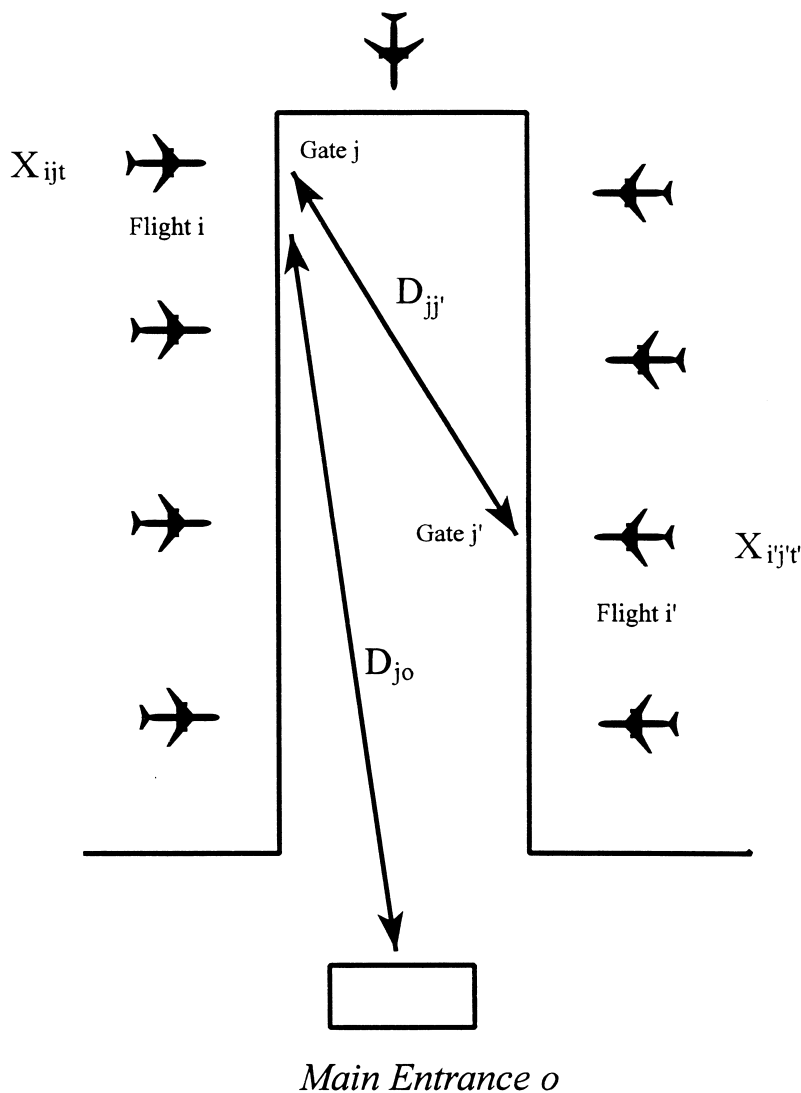


Fig. 2. A simple terminal configuration.

$$Y_{ijj'} = \begin{cases} 1 & \text{if flight } i \text{ is assigned to gate } j \text{ and flight } i' \text{ is assigned to gate } j' \\ 0 & \text{otherwise} \end{cases}$$

Now, the QAP is equivalent to the integer linear program of the form:

$$\text{MIN} \sum_i \sum_{i'} \sum_j \sum_{j'} P_{ii'} D_{jj'} Y_{ijj'} \quad (7)$$

and it is subject to several proper constraints. Let F be the number of flights in the schedule, G be the number of gates, T be the number of time intervals, and T_i^o be the total time intervals that flight I dwells on the ground. Then, the first class of constraints is necessary for the flight-to-gate assignment to meet the following two conditions.

1. Every flight must be assigned to one and only one gate during its apron time:

$$\sum_j X_{ijt} = 1 \quad \forall i, t_i^a \leq t \leq t_i^d \quad (8)$$

where t_i^a is the arrival time period of flight I and t_i^d is the departure time period of flight I . Also, the total time intervals of flight I dwell on the ground T_i^o is equal to the difference

between t_i^a and t_i^d . The number of such constraints is equal to the number of flights time the total dwell time intervals for each flight, $FXT_i^a = F^*(t_i^d - (t_i^a - 1))$

2. No two aircraft may be assigned to the same gate when their apron times overlap:

$$\sum_i X_{ijt} \leq 1 \quad \forall j, t \quad (9)$$

There are $G \times T$ constraints of this type for G gates and T time intervals.

The objective function eqn (5) subject to constraints eqns (8) and (9) without time interval conditions is a conventional QAP. Now, we add two more conditions to meet the time constraints which correspond to the time tables of flight schedules.

3. Each aircraft must be assigned to the same gate during its apron time:

$$X_{ijt} \leq X_{ij(t+1)} \quad \forall t, t_i^a \leq t \leq t_i^d - 1 \quad (10)$$

4. $Y_{ij't'j'} = X_{ijt}X_{t'j't'}$ can be written as a linear constraint:

$$X_{ijt_i}^a + X_{ijt_t}^a - 2Y_{ij't'j'} \geq 0 \quad \forall i < i', j, j' \quad (11)$$

Conversely, let X_{ijt} , $Y_{ij't'j'}$ be a feasible solution of objective function eqn (7). From $X_{ijt} + X_{i't'j't'} - 2Y_{ij't'j'} \geq 0$, $X_{ijt} \in \{0, 1\}$, $Y_{ij't'j'} \in \{0, 1\}$, it follows that $Y_{ij't'j'} = 1 \Rightarrow X_{ijt} = X_{i't'j't'} = 1$. If we assume $X_{ijt} = X_{i't'j't'} = 1$, but $Y_{ij't'j'} = 0$, we have a contradiction to the following constraint.

5. Conservation constraints:

$$\sum_j \sum_{j'} Y_{ij't'j'} = 1 \quad \forall i \neq i' \quad (12)$$

Therefore $Y_{ij't'j'} = 1 \Leftrightarrow X_{ijt} = X_{i't'j't'} = 1$, which can be written as $Y_{ij't'j'} = X_{ijt}X_{i't'j't'}$. Thus the objective function eqn (7) yields the same value as that of eqn (5).

The total number of constraints in the integer program generated by the above formulation is $F^*T_i^o + G^*T + F^*G^*T_i^o + F^*(F-1)^*G^*G + F^*(F-1)$ for F flights, G gates and T time intervals, which also includes $F^*(F-1)^*G^*G + 2^*F^*G^*T$ integer variables in the formulation. For example, if there were 5 flights, 4 gates, and 7 time intervals in the same day's schedule, the total number of constraints would be 432 and 480 integer variables accordingly. However, many of these constraints are redundant.

Finally, a proper linear integer program, objective function eqn (7) subject to constraints eqns (8)–(12), is formulated for the new gate assignment problem. The important difference between the new formulation of GAP and the conventional formulation of GAP is that the new formulation brings in the time constraints to deal with the situation when flight schedules are overlapping during the study time periods. Here, we can use a branch-and-bound technique to solve this linear integer programming formulation.

4. A HEURISTIC APPROACH

The burden of long walking distances on air travelers can also be eased by implementing simple, yet efficient, assignment procedures. For example, assign the best available gate to the first arriving flight. Many heuristic algorithms like this were tested (Mangoubi, 1985), and as a result one heuristic showed the best performance overall in reducing walking distances. The heuristic consists of assignment of the aircraft with the greatest number of on-board passengers to the gate with the corresponding shortest total walking distances. However, this heuristic doesn't consider the flight schedules over a long time period or a successive time period. One possible gate assignment for successive flights might tend to park at the same gate in order to minimize the total passenger

walking distances when the flight schedules aren't overlapping, and this assignment won't be included in the conventional problem.

Similarly, Chang (1994) proposed several algorithms to solve the QAP as a GAP, such as the accessibility value sorting, the Sorting-Los, and Sorting-Mixed algorithms. For the purpose of busy hub operations, these algorithms also perform well in efficiently solving the GAP for a batch of aircrafts arriving at a hub. However, these algorithms are still unable to apply to the GAP with the time constraints. It is not easy to just exchange any two or three flight-to-gate combinations to improve the walking distances by using any exchange algorithm. Therefore, we propose a new heuristic algorithm to extend the work on dealing with the GAP with time constraints.

First, the proposed heuristic algorithm tends to assign the successive flights parking at the same gate when the flight schedules are not overlapping. We can easily understand that only one gate is needed when all flight schedules are not overlapping to each other and we also obtain the minimum total walking distances in this special case. This is not possible all the time. Second, if flight schedules are overlapping, we list the decision variables, $Y_{iji'j'}$, in increasing order of their objective function coefficient and then we begin to assign the flights to available gates based on that the first variable $Y_{iji'j'}$ with the smallest cost coefficients ($P_{ii'}^* D_{jj'}$), or the shortest total walking distance, is equal to one in the list. When a set of solutions meet all requirements of gate assignment which includes that all flight are assigned and do not violate flight schedules, one set of feasible solutions has been obtained and label variable $Y_{iji'j'}$, then redo the whole processes again based on the next unlabelled variable $Y_{iji'j'}$ in the list. Therefore, we might obtain several sets of feasible solutions and then we will choose the best solutions for the minimum walking distances. The proposed algorithm proceeds as follows:

- Step 0.* *Initialization.* Based on the flight's time table, we can eliminate all decision variables $Y_{iji'j'}$ with conflicting gate assignment. For example, flight I 's dwell time overlaps the flight j 's dwell time, then we can eliminate (mark) all variables $Y_{iji'j'}$ corresponding flight I and j dwelling at the same gates, which can reduce the number of integer variables considered in the heuristic algorithm. Note that at this stage any incompatible flight-gate pair can also be eliminated from further consideration.
- Step 1.* *Sorting the decision variables.* List the decision variables, $Y_{iji'j'}$, in increasing order of their objective function coefficient which are the transferred passenger walking distances between two possible transferred gates assignment. Then, label these variables from $n = 1$ to M .
- Step 2.* Let $n = 1$
- Step 3.* Pick up the n th feasible variable $Y_{iji'j'}$ in the list and set $Y_{iji'j'} = 1$ to be the feasible solution in set A.
- Step 4.* Let $K = n + 1$
- Step 5.* Check the k th variable, if the k th variable has not been labelled, examine if this variable is feasible. Check if the variable corresponding to the schedule and gate assignment is conflicting with the variables in set A; if yes, set $Y_{iji'j'} = 1$ and put into set A; if no, label this variable. If the k th variable has been labelled, go to step 6.
- Step 6.* Check if $K = M$, if yes, go to step 7. Otherwise, let $K = K + 1$ go to step 5.
- Step 7.* The feasible solution set will be the best solution so far. Then, check if $n = M$, if yes, stop; otherwise, reset the set A to 0 and let $n = n + 1$, go to step 3.
- Step 8.* Check all of the feasible solution sets and choose the best solution which has the best total walking distances.

The flow chart of this algorithm is shown on Fig. 3.

5. EMPIRICAL STUDY

Modern airport terminal building designs have finger piers, which are linear or satellite. For example, hub airports, with a large proportion of transfers especially, have different needs from local airports, with a majority of originating and terminating traffic.

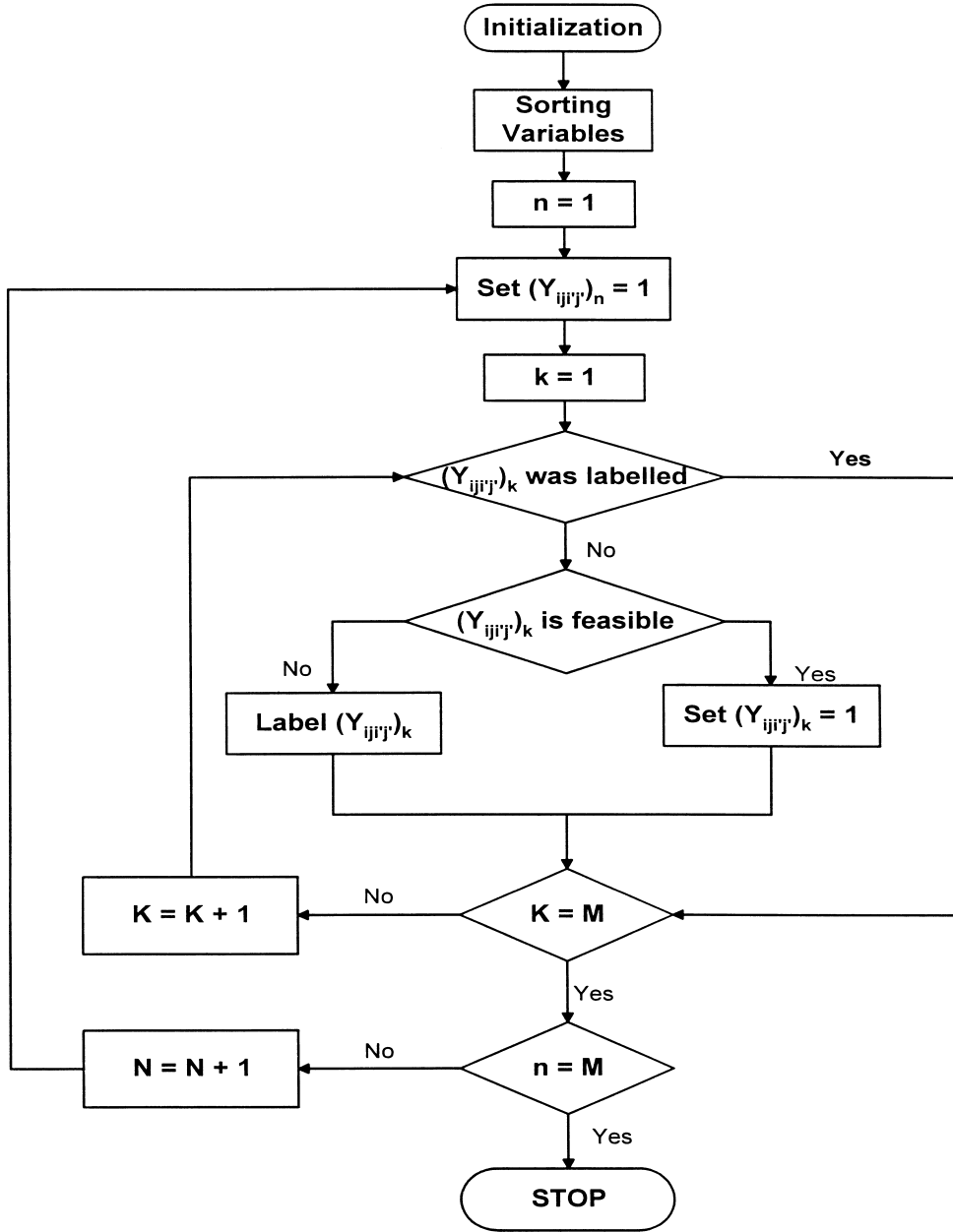


Fig. 3. The flow chart of proposed heuristic algorithm.

This suggests different 'ideal' layouts. This was first noted by De Neufville (1976). As proposed by De Neufville and Rusconi-Clerici (1978), passenger-terminal design should depend not only on the traffic, but on the 'composition' of the traffic. Although the proportion of transferring passengers in a major hub can be around 50% (Chicago, Dallas, Denver, Miami) and as much as 70% (Atlanta), for the most part airport planning studies have not considered transferring passengers. This 'design oversight' could arise from the difficulty in forecasting traffic composition, with the changing conditions in a competitive market. Regarding the two terminal concepts, linear and pier, some empirical studies conclude that a terminal with pier-finger configuration is better than the linear terminal (regarding walking distance) when the percentage of transfers exceeds 30% (De Neufville and Rusconi-Clerici, 1978). These studies also show an equivalence in the United States between the proportion of 'on-line' transfers (within the same airline) and the total proportion of transfers at the airport.

In this study we considered three types of terminal configurations: linear type with one sided parking, linear type with two sided parking, and pier type, as shown in Figs 4, and 5. Also, we

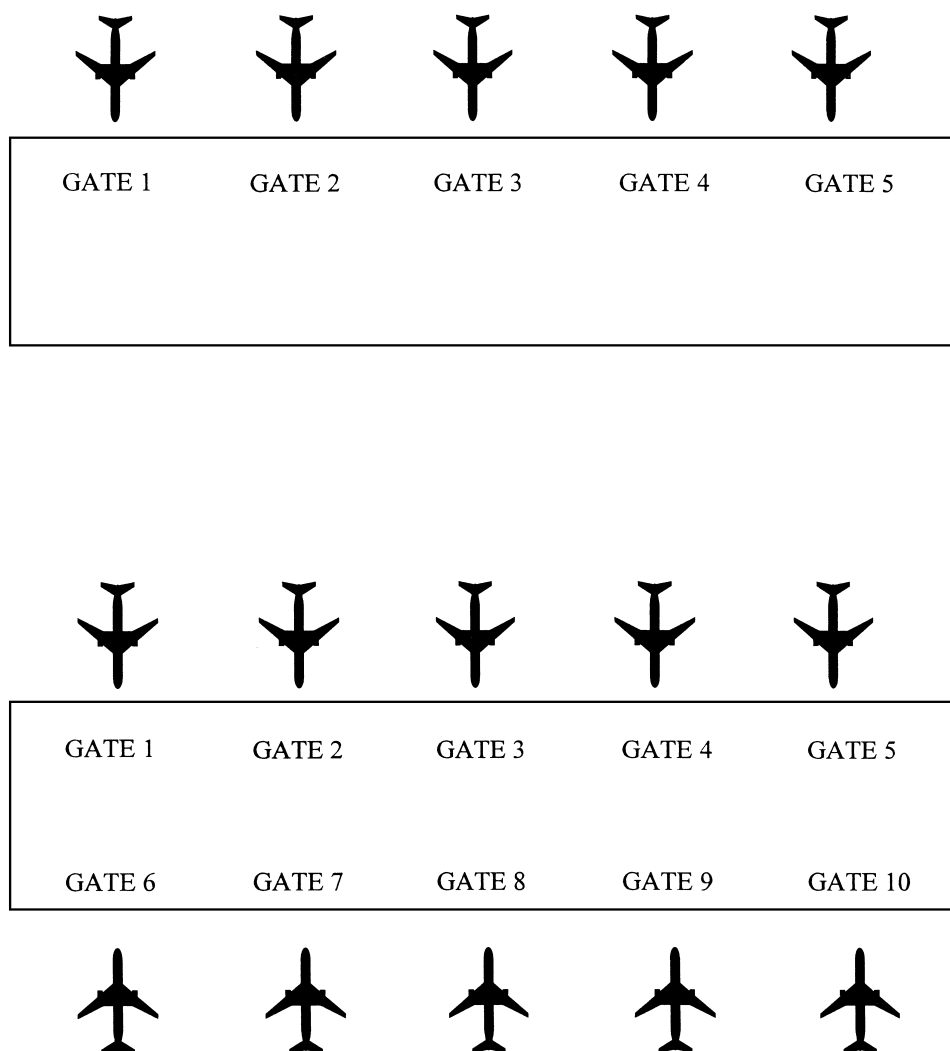


Fig. 4. One-sided and two-sided parking terminals.

assumed the distances in both directions between any two adjacent gates are equal in linear types of terminal configurations. However, the pier type terminal may have different walking distances between two neighbor gates because walking patterns have differences in order to reduce conflict points when passenger cross the terminal to reach the gates on the other side. The matrices of passenger traffic flow (passenger O–D tables) are randomly generated. All problems have different O–D tables associated with different numbers of flights.

We present the computation results obtained with the heuristic approach and those obtained from directly solving the IP formulation. Since the GAP is an NP-hard problem (Obata, 1979), it is quite difficult to solve the IP formulation in a reasonable execution time using commercial IP solver software, such as LINDO, CPLEX, etc. So, we were not able to have all IP solutions for each set of problems. This computation study is intended to verify the effectiveness of the proposed heuristic algorithm for the minimum total walking distances in gate assignment. The proposed heuristic algorithm is coded in C language.

There are three sets of the test problems, each of which contains several problems. The first problem set was taken from Nugent *et al.* (1968) and Obata (1979). The problems in this set are used to compare the quality of the solutions obtained using the proposed heuristic algorithm with the existing models. Problems here include no more than 10 gates and the gate configuration is linear. Also, there is only one time interval in each problem. This is the conventional gate assignment problem for a batch of arriving aircrafts during one time period. There are five different problem sizes (in terms of the number of aircrafts or gates), i.e. 5, 6, 8, 9, and 10. We did not test

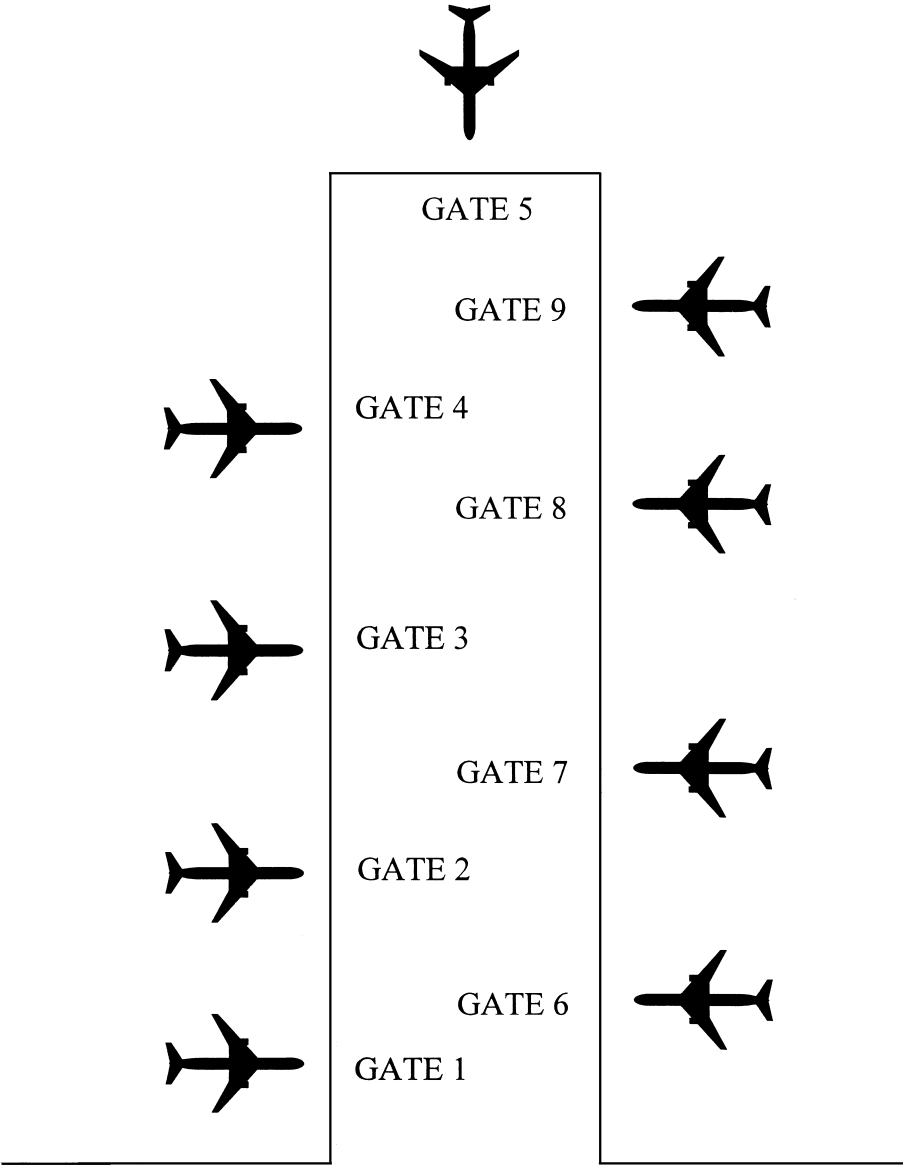


Fig. 5. A simple pier-finger terminal.

larger problem sizes because the comparison between proposed heuristic algorithm and existing algorithms for the conventional gate assignment problem was not the major goal of this study. However, the conventional gate assignment problem can be considered a special case of the more general gate assignment problem which we intended to solve.

Table 1 displays the first feasible solutions and the final solutions obtained using the proposed heuristic algorithm. Table 2 displays the comparison between the final solutions of heuristic

Table 1. Objective function values for heuristic algorithm

Problem	No. of flights	1st feasible solution	Final solution	% of Improvement
1	5	190	150	21.05
2	6	380	292	23.16
3	7	638	482	24.45
4	8	928	784	15.52
5	9	1280	1034	19.22
6	10	1818	1436	21.01
7	10	782	422	46.04
8	10	1460	740	49.32
9	10	2718	1623	40.29

Table 2. Comparison between final solution of heuristic algorithm and best solution from literature

Problem	No. of flights	Heuristic	Best solution in literature	Source
1	5	150	150	Obata
2	6	292	292	Obata
3	7	482	482	Obata
4	8	784	784	Obata
5	9	1034	1034	Obata
6	10	1436	1436	Nug.
7	10	422	422	Obata
8	10	740	740	Sury.
9	10	1623	1623	Sury.

algorithm and best known solutions from the literature for the problem under consideration. In all cases considered, the proposed solution approach obtained the best known solutions which have been reported in the literature. Therefore, for these cases, the proposed heuristic approach performed at least as well as other existing approaches for the gate assignment problem.

The second set of problems were artificially generated problems. These problems were used to compare the solutions from proposed heuristic algorithm and those obtained using commercial software, CPLEX. The problems here included no more than eight flights and four gates because CPLEX for the personal computer version could not handle more than 1800 integer variables in these problems without extended memory. There are three different problem sizes (number of aircraft), i.e. 6, 7, and 8 flights and 4 gates for all problem sizes. In addition, 6 and 7 flights were scheduled during 8 time periods and 8 flights were scheduled for 9 time periods. We tested five cases for each problem size. Each case had a different flight schedule which was randomly planned.

Table 3 displays the first feasible solution and final solution obtained using the proposed heuristic algorithm.

Since there is no guarantee that the proposed heuristic procedure will produce the optimal solution, we report the 1st feasible solutions as well as the final solutions to show that the heuristic approach does in fact improve upon its initial solution as it proceeds through several iterations. Table 3 also indicates that in all cases examined, the proposed solution approach produced the optimal solution as its final solution, and the execution time for CPLEX is much greater than the execution time for the proposed heuristic. When the number of flights increases, the execution time using CPLEX increases more rapidly. So, although CPLEX is capable of solving larger problems, its use is impractical due to significant required computational time.

Table 4 compares the number of iterations required to obtain the best solution and the corresponding execution times to the total number of iterations and the corresponding execution times for all of these cases. It is obvious from Table 4 that in all cases considered, the best solutions are

Table 3. Objective function values from heuristic algorithm and CPLEX software

Case	Heuristic				CPLEX	
	1st feasible	Final solution	Improvement (%)	Running time (s)	Optimal solution	Running time (s)
C6481	229,550	210,000	8.52	4.18	210,000	871.60
C6482	231,150	220,425	4.64	3.68	220,425	491.01
C6483	221,200	220,200	0.45	3.74	220,200	205.37
C6484	229,550	219,150	4.53	3.79	219,150	405.62
C6485	226,850	220,425	2.83	3.63	220,425	202.95
C7481	44,600	41,310	7.38	8.41	41,310	3776.84
C7482	47,460	46,720	1.56	7.31	46,720	325.63
C7483	44,180	44,180	0.00	7.42	44,180	840.08
C7484	49,360	45,200	8.43	8.35	45,200	2422.20
C7485	49,880	43,520	12.75	7.42	43,520	431.33
C8491	72,300	61,200	15.35	14.12	61,200	1390.22
C8492	73,050	53,580	26.65	19.29	53,580	12592.6
C8493	64,680	60,800	6.00	16.43	60,800	876.12
C8494	67,380	65,820	2.32	16.76	65,820	1556.98
C8495	75,400	54,720	27.43	17.36	54,720	1616.46

Table 4. Performance of heuristic algorithm for comparison cases

Case	No. of integers	Final solution	Best soln. iteration no.	Running time for best soln. (s)	Total no. of iterations	Total running time (s)
C6481	480	210,000	17	2.25''	199	4.18
C6482	480	220,425	45	1.43''	103	3.68
C6483	480	220,200	2	0.66''	74	3.74
C6484	480	219,150	17	1.15''	109	3.79
C6485	480	220,425	31	1.26''	76	3.63
C7481	672	41,310	18	1.32''	239	8.41
C7482	672	46,720	15	1.37''	64	7.31
C7483	672	44,180	1	0.93''	78	7.42
C7484	672	45,200	29	1.76''	215	8.35
C7485	672	43,520	13	1.21''	93	7.42
C8491	896	61,200	19	4.12''	71	14.12
C8492	896	53,580	53	3.74''	316	19.29
C8493	896	60,800	32	7.75''	37	16.43
C8494	896	65,820	12	5.00''	55	16.76
C8495	896	54,720	33	3.90''	113	17.36

obtained in relatively few iterations compared to the total number of iterations which are required for the algorithm to stop. In fact in all cases, the execution time to obtain the best solution is significantly less than half of the total execution time. This implies that the search loops on the proposed heuristic approach can be stopped after a certain number of iterations and still one can be confident that a good feasible solution is in hand. More cases need to be examined to confirm this assertion.

The third set of problems were used to test the performance of the proposed heuristic algorithm in realistic problem contexts. The problems here range from small to large problems. There are 15 different problem sizes. Also, the flight schedules for each size problem are randomly planned. All of the test problems were run on the 486DX-33MHz PC DOS-based computer except for cases 17 and 18. Table 5 lists the problem sizes and displays the first feasible solutions and the final solutions obtained using the proposed heuristic algorithm and the total execution time. The final solutions in cases 1–8 are the same as the optimal solutions which were obtained using CPLEX software. Cases 17 and 18 were tested in a UNIX-based environment to circumvent memory problems.

Table 6 displays the performance of the heuristic algorithm. The second column in the table lists the number of integer variables ($Y_{iji'j'}$) for each problem size, which indicates the largest number of loops may be searched for the feasible solutions by applying the proposed heuristic searching rules. However, there are many variables which are redundant because of overlapping flight schedule. The proposed heuristic will eliminate these redundant variables and the searching loops will be less than the largest loops. In addition, the relationship between the execution time of the best

Table 5. Objective function values for heuristic algorithm

Case no.	No. of flights	No. of gates	No. of intervals	1st solution	Final solution	% of improvement	CPU time
1	3	2	6	52,750	37,350	29.19	0.11''
2	5	3	7	193,100	173,750	10.02	0.49''
3	5	4	7	143,950	142,700	0.87	1.37''
4	5	4	8	170,800	138,850	18.71	1.43''
5	6	4	8	229,550	210,000	8.52	3.41''
6	6	4	8	231,150	220,425	4.64	2.97''
7	7	4	8	44,600	41,310	7.38	6.92''
8	8	4	9	72,300	61,510	14.92	11.92''
9	9	4	9	91,090	80,460	11.67	22.25''
10	9	5	9	74,380	67,640	9.06	52.69''
11	10	5	9	75,060	50,120	33.23	1'39''
12	11	5	9	72,600	48,860	32.70	2'38''
13	15	5	13	253,740	212,470	16.26	10'1''
14	16	5	13	458,500	231,500	49.51	18'1''
15	17	5	14	276,230	261,490	5.34	20'1''
16	18	5	14	314,760	284,160	9.72	27'5''
17	20	5	15	404,510	404,510	0.00	11'47''
18	30	7	20	738,630	690,580	6.51	112'2''

Table 6. Performance of heuristic algorithm for comparison cases

Case	No. of integers	Final solution	In which no. of feasible soln.	Running time for best soln. (s)	Total no. of feasible soln.	Final running time (s)
1	24	37,350	2	0.05	3	0.11''
2	180	173,750	3	0.16''	40	0.49''
3	320	142,700	114	1.32''	124	1.37''
4	320	138,850	110	1.26''	132	1.43''
5	480	210,000	117	2.25''	199	3.41''
6	480	220,425	45	1.43''	103	2.97
7	672	41,310	18	1.32''	239	6.92''
8	896	61,510	19	4.12''	71	11.92''
9	1152	80,460	4	3.74''	55	22.25''
10	1800	67,640	5	8.90''	148	52.69''
11	2250	50,120	145	40.07''	300	1'39''
12	2750	48,860	9	20.43''	258	2'38''
13	5250	212,470	12	36.45''	65	10'1''
14	6000	231,500	37	1'38''	56	18'1''
15	6800	261,490	15	1'41''	32	20'1''
16	7650	284,160	2	44.16''	42	27'5''
17	9500	404,510	1	1'12''	12	11'47''
18	42,630	690,580	4	30'20''	29	112'2''

solution and the total execution time of each case by using the heuristic algorithm is still quite different. For most of cases, the execution time of the best solution is less than a half of the total execution time for a large size problem.

When the number of gates is greater than 15, we can not obtain an optimal solution in a reasonable time. Figure 6 shows when the problems have more than 2250 integer variables ($Y_{iji'j'}$) in terms of flight-to-gate possibilities, the average execution time is increasing rapidly. In addition, the execution time of the best solution for several cases has a linear relationship, which may imply that the proposed algorithm is a good heuristic approach. In general to handle larger problems with more flights and gates, one approach could be to decompose the problem into several smaller ones. For examples, the gate assignment in different terminals can be solved as independent problems. In general, to plan for gate assignments which is done infrequently, one can afford to solve a very large problem even if it takes a large amount of computation time.

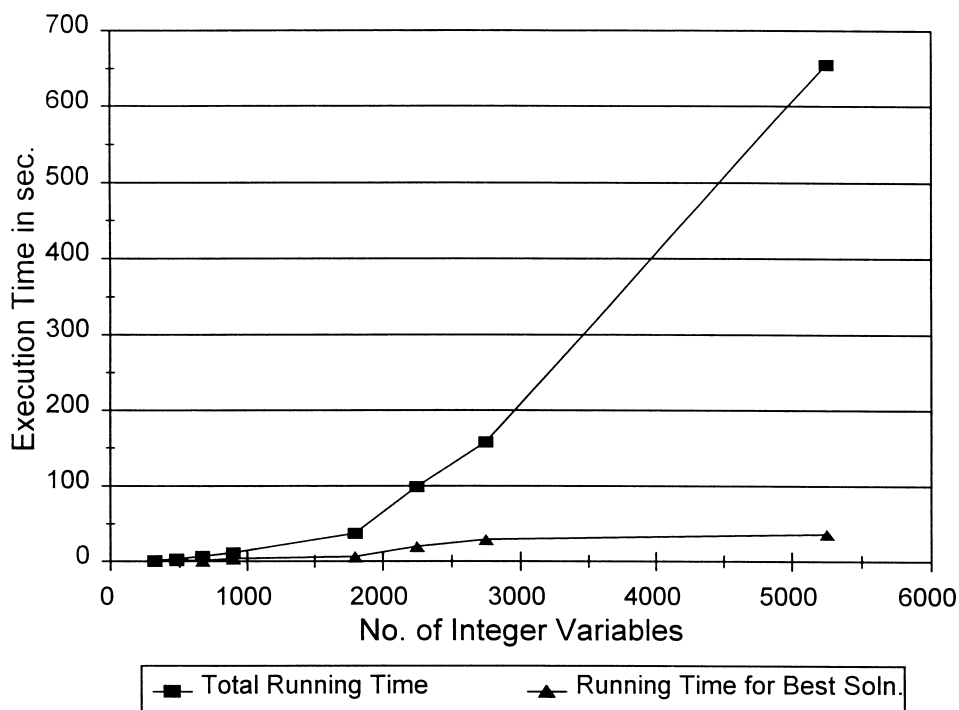


Fig. 6. Execution time of heuristic algorithm for different problem sizes.

Based on above analysis, several observations can be made. First, the proposed heuristic algorithm obtained good solutions for the conventional GAP on several small problem sizes. Second, in all cases where comparisons were made with the CPLEX solutions, the best solutions obtained from the proposed heuristic algorithm were the optimal solutions as obtained by CPLEX software. Third, considering the required execution times and the problem sizes, the proposed heuristic approach performs very well in solving the GAPs. Finally, in all cases considered, the execution times required to obtain the best heuristic solution displays a linear relationship with the size of the problem in terms of the number of integer variables which implies that the proposed algorithm is an efficient heuristic approach.

6. CONCLUSIONS AND DIRECTIONS FOR FUTURE RESEARCH

A new gate assignment problem which minimizes passenger walking distances based on flight schedules is formulated and solved. It differs from the conventional GAP, which focuses on minimizing passenger walking distance during the arrival of a batch of aircrafts, by incorporating the time dimension in the problem. The GAP is formulated here as binary integer QAP. This implies that the gate assignment problem considered is an NP-hard problem. Due to the complex nature of the problem, a heuristic algorithm was proposed to solve the GAP efficiently.

Three sets of the test problems, each containing several problems are solved. All passenger flows are randomly generated. The results show that the proposed heuristic algorithm did obtain the best solutions when compared to the results from the literature when the problem sizes were less than 10 flights and 10 gates.

In addition, we compared the solutions obtained from CPLEX software in order to examine the quality of solutions obtained from the of heuristic algorithm. The results of these tests showed that the heuristic algorithm obtained the best solutions for all examples. We also examined some large problems which involved more than 10 flights and 5 gates. When the number of flights and gates increased, the integer variables also increased rapidly. Therefore, the largest problems involving 20 flights and 5 gates, and 30 flights and 7 gates are solved in an UNIX environment. Although in the large problem cases we do not have an optimal solution to compare, based on the significant improvements between the initial and final solutions obtained using the proposed method, and the performance of the approach in other cases we can conclude that the proposed approach is an efficient approach which provides close to optimal solutions for a very complex problem.

The research presented in this paper can be extended in several avenues. The proposed formulation and the solution procedure are useful for gate assignment at the planning level. The model and the algorithm can be extended to deal with the real-time GAP. This is important when due to unforeseen delays, the planned gates have to be re-assigned in real time.

A more comprehensive baggage transport cost function and manpower, equipment, and vehicle constraints may be worth pursuing for estimating total cost more realistically. A baggage handling cost function may be estimated with the least squares method if the field data are available. The loading, transport, sorting, and unloading times should be functions of the number of the available operators, carts, and vehicles. If the comprehensive baggage handling cost function and manpower, equipment, and vehicle constraints are successfully incorporated in the current model, a more realistic total cost and gate assignments can be estimated.

When the GAP is solved more effectively, then alternative gate assignment plans can be evaluated more rapidly in light of the real world operational constraints which are not well represented in the problem formulation.

The flight scheduling problem can be re-examined to evaluate whether or not the current flight schedule is suitable for the current airport operations or vice versa. Although we cannot use the model presented in this study to solve the flight scheduling problem, we can examine the implications of alternative flight scheduling in airport operations.

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